

INSTRUCTOR PACKET

Metabo-Diet

*Harmonizing Dietary and Exercise Phenotypes
with Metabolomics Across CFDE Resources*

Facilitation, assessment, and scientific-safety resources

Release 1.0 | August 2026

Instructor-only: keep assessment keys hidden until learner submission

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Metabo-Diet instructor guide

Module at a glance

Title: Metabo-Diet: Harmonizing Dietary and Exercise Phenotypes with Metabolomics Across CFDE Resources

Level: Intermediate

Instructional time: 140 minutes

Primary format: Guided mini-lessons, structured worksheets, and an interactive Python notebook

Hands-on data source: Public Metabolomics Workbench studies only

Design framework: Backward design with Bloom progression from understand to apply, analyze, evaluate, and create

This module teaches learners to decide what the two studies can support. Keeping an incompatible field, feature, matrix, timepoint, or quantitative table separate is an acceptable result.

Scope: The module compares metadata structure and reviewed analyte-name coverage and performs quantitative exploration within each study. It does not pool quantitative values across the two studies or estimate a diet-versus-exercise causal effect.

Audience and prerequisites

The module is intended for graduate students, postdoctoral scholars, research staff, and analysts in metabolomics, nutrition, exercise biology, translational omics, or computational biology.

Learners should be able to:

- Read a rectangular data table.
- Read guided Python and pandas cells and change the named values when prompted.
- Interpret a scatterplot.
- Recognize samples, variables, and missing values.
- Explain at a basic level what metabolomics measures.

No prior Metabolomics Workbench, RefMet, MoTrPAC, NPH, or All of Us experience is required.

No prior terminal or Jupyter experience is required for the guided path, but learners need basic familiarity with tables and scatterplots. The module is not a general Python course. Direct first-time setup appears before the pretest; R is optional.

Learning objectives

By the end of the module, learners will be able to:

- **LO1:** Identify and compare intervention structure, specimen context, timepoint design, and clinical or behavioral phenotype representation across the two Metabolomics Workbench studies.
- **LO2:** Describe how study design, metadata provenance, analytical platform, and preprocessing influence what can and cannot be concluded from cross-study comparison.
- **LO3:** Apply a RefMet-centered workflow that aligns analyte names, specimen and time semantics, variables, and units while recording field-level decisions.
- **LO4:** Retrieve and analyze public MW data through the REST API and interpret overlap, class summaries, and within-study PCA in context.
- **LO5:** Evaluate the current access pattern of a candidate resource and adapt retrieval, compute, storage, cache, and output architecture accordingly.

Objective-instruction-assessment alignment

Objective	Instruction and practice	Embedded evidence	Summative evidence
LO1	Lessons 1-2; cohort-comparison worksheet; sample/time audit in Lesson 4	KC2-01 to KC2-03; sourced compatibility decisions	POST-01 to POST-03; completed comparison worksheet
LO2	Lessons 1-4; interpretation limits; bounded-claim exercise	KC1-01, KC1-03, KC2 items, KC3-02, KC4-03	POST-02 to POST-04, POST-08; four-sentence interpretation
LO3	Lesson 3; crosswalk and overlap audit	KC3-01 to KC3-03; reviewed ambiguous/unmapped rows	POST-05 to POST-07, POST-12; crosswalk rubric
LO4	Lesson 4; REST retrieval, audit, overlap, class summary, PCA	KC4-01 to KC4-03; notebook audit outputs	POST-07 to POST-09, POST-12; reproducible notebook execution
LO5	Lessons 1 and 5; transfer checklist	KC1-02 and KC5-01 to KC5-03	POST-10 and POST-11; sourced go/revise/wait/stop decision

The pretest contains at least two items per objective domain when cross-listed items are included. Use objective-level change, not only total score, when evaluating module value.

Instructional schedule

Lesson	Minutes	Main learner artifact
1. Why phenotype-to-metabolome harmonization matters	20	Pretest, access evidence, reflection
2. Comparing study design and phenotype capture	30	Cohort-comparison worksheet
3. Harmonizing metabolomics and metadata	35	Metabolite/metadata crosswalk
4. Guided analysis and biological interpretation	40	Retrieval audit, overlap flow, class summary, within-study PCA, bounded interpretation

Lesson	Minutes	Main learner artifact
5. Access patterns and transfer	15	Transfer checklist and posttest
Total	140	

The 140 minutes include embedded knowledge checks but exclude the 5-minute pretest, 8-minute posttest, and software installation, for an expected course total of about 153 minutes after setup. If learners are new to Python, Jupyter, PCA, or metabolomics, schedule 30 to 60 additional minutes and use the reduced worksheet path stated in each lesson.

Locked data configuration

```
DIET_ACCESSION = ST001 521
EXERCISE_ACCESSION = ST003 348
```

Diet study facts to verify before delivery

- Human plasma FARMM study with three deposited diet labels: [Vegan](#), [Western](#), and [Modulen](#).
- Study summary describes 30 volunteers and five biological timepoints.
- Current factors response contains 160 rows: 150 participant-timepoint rows and 10 [QPP...](#) pooled QC candidates.
- Longitudinal time course includes an antibiotic intervention on days 6-8 and polyethylene glycol on day 7.
- Vegan participants were established vegans and remained outpatients; 20 omnivores were randomized to omnivore versus EEN in an inpatient setting.
- Four analyses: AN002533 through AN002536, reporting unitless peak areas.

Exercise study facts to verify before delivery

- Human endurance race-walking study with 19 athletes represented at four timepoints and 76 samples.
- [REST](#): pre-exercise; [STAT](#): immediately post-exercise; [REC3](#): 3-hour recovery; [REC22](#): 22-hour recovery.
- Factors endpoint reports broad source [blood](#); collection metadata specifies serum and describes clotting and processing.
- Two analyses: AN005483 and AN005484, reporting peak area.

Overlap audit targets

- 153 shared nonblank exact RefMet strings before artifact and eligibility exclusions.

- Eight [ST003348](#) isotope/internal-standard rows map to ordinary RefMet labels and are excluded from biological overlap.
- 145 conservative biological overlaps after that filter.

These are versioned reproducibility checks, not scientific constants. If live data differ, investigate version, selected analyses, blank handling, duplicate collapse, and artifact rules. Do not change logic simply to force the expected values.

Reserve pair

If a locked study is withdrawn or no longer usable, the recommended reserve pair from the scoping audit is:

```
DIET_ACCESSION = ST000292
EXERCISE_ACCESSION = ST001789
```

Both are plasma candidates, and the scoping audit found 122 raw exact-name overlaps. [ST000291](#) is the urine companion to the diet project and should not be substituted for [ST000292](#) without redesigning the specimen comparison. Before activation, rerun all release, metadata-completeness, overlap, sample-role, and licensing checks and generate a new provenance manifest. Never switch pairs silently during a cohort.

Instructor setup checklist

Complete within 48 hours of delivery:

- [] Open the current [MW REST documentation](#).
- [] Confirm both study landing pages and summary, factors, analysis, metabolites, and data endpoints are public.
- [] Record UTC retrieval times and license strings.
- [] Run the notebook from a fresh environment with live retrieval.
- [] Confirm the first-time setup works before opening the pretest and that the smoke-test command writes a separate notebook.
- [] Verify structural counts, [QPP...](#) sample roles, analysis IDs, and plasma-versus-serum evidence.
- [] Verify the 153 raw and 145 conservative overlap audit stages.
- [] Test the cached fallback by simulating live-retrieval failure without overwriting the live cache.
- [] Confirm cache checksums and visible live/cached labels.
- [] Confirm no credentials, tokens, signed URLs, governed data, or personal information appear in materials or outputs.
- [] Validate all JSON assessment files.
- [] Check every external link and label any time-sensitive access claim with a date.

- [] Review MoTrPAC and All of Us first-party access documentation for changes.
- [] Test keyboard-only lesson, form, and notebook navigation.
- [] Test headings, table headers, link labels, figure descriptions, and color contrast with accessibility tooling.
- [] Ensure answer rationales are hidden until submission.

Facilitation guidance by lesson

Lesson 1

Emphasize the difference between common retrieval, structural harmonization, and statistical exchangeability. Ask learners to identify a familiar situation where a shared column label hid a protocol difference. Do not introduce the module as a method for combining cohorts; introduce it as a method for making a defensible comparability decision.

Current-access correction: the July proposal described MoTrPAC as registration-gated. Current MoTrPAC documentation distinguishes public and restricted pathways. Explain that the module intentionally updates the proposal because access is a time-stamped dataset property.

Lesson 2

Have learners extract each study independently before comparing columns. Watch for three common errors:

1. Calling 160 factor rows in [ST001 521](#) participants.
2. Calling all three diet groups randomized.
3. Calling the exercise matrix simply blood and overlooking the serum-specific collection block.

Treat a documented [not reported](#) or [not yet assessable](#) as better work than an unsupported completed cell.

Lesson 3

Reward auditability rather than mapping rate. Require at least one ambiguous, one unmapped, one many-to-one, and one artifact or nonbiological example when present. The [Palmitic acid-\[13C\]16](#) to [Palmitic acid](#) example illustrates why a nonblank RefMet field still requires source-role review.

Do not allow learners to overwrite the submitted name or discard blank mappings. Ask what analysis each eligibility flag serves.

Lesson 4

Run retrieval and validation before any figure. Pause if join cardinality increases rows. Keep pooled QC samples available for assay diagnostics and excluded from biological PCA. Fit PCA separately within studies; the release module intentionally omits a combined PCA of the uncalibrated raw matrices.

When discussing the overlap, say:

153 tests the raw nonblank exact-string step; 145 tests the conservative exclusion of eight isotope/internal-standard mappings. Neither number measures a shared intervention response.

Require the four-sentence interpretation scaffold: observation, context, alternatives, boundary.

Lesson 5

Require first-party, dated evidence for the exact dataset and action. A valid transfer decision may be [WAIT](#) or [STOP](#). Correct the claim that no output can ever leave All of Us: participant-level data remain governed, while only outputs permitted under current dissemination policy may leave.

Assessment administration

Pretest

- Administer before Lesson 1 content.
- Do not show answers until submission.
- Use as a baseline and diagnostic, not as a gate.
- Maximum: 10 points.

Embedded knowledge checks

- Use immediate feedback with rationale after each response.
- Permit retry after the rationale.
- Record first-attempt responses for instructional evaluation if the platform and consent permit.

Posttest

- Administer after the transfer activity.
- Maximum: 12 points.
- Suggested mastery: 10/12, with no automatic certificate based on total score alone.
- Learners missing both items in an objective domain should review the associated lesson and complete a parallel retry.

Performance artifacts

Knowledge items do not fully assess application. Require these artifacts for completion:

- Cohort-comparison worksheet meeting all completion checks.
- Crosswalk scoring at least 16/20 and meeting all safety conditions.
- Notebook audit reproducing or explaining the structural and overlap checkpoints.
- A bounded four-sentence interpretation with no causal cross-study claim.

- Transfer checklist scoring at least 16/20 and meeting all minimum pass conditions.

Accessibility guidance

The lesson Markdown includes text descriptions for conceptual access, comparison, class-summary, and PCA figures. Preserve those descriptions when converting to dashboard components.

Content requirements

- Use semantic heading levels in order.
- Give tables real header cells and avoid merged cells where possible.
- Do not rely on color alone; pair color with shape, line type, or direct labels.
- Use descriptive links rather than [click here](#).
- Expand abbreviations on first use and maintain the glossary link.
- Provide a text table behind every chart with the plotted values and sample counts.
- Include explained variance and figure-specific axis text for each PCA.
- Label the separate figures explicitly as plasma/diet and serum/exercise.
- Avoid motion or require a pause control.

Notebook requirements

- Every code cell needs a preceding plain-language purpose statement.
- Every output needs a short interpretation and a machine-readable or text alternative.
- Errors must be conveyed in text, not color alone.
- Interactive widgets need keyboard focus, labels, and a non-widget fallback.
- Figures need high-contrast palettes and marker shapes.
- Do not auto-scroll past warnings or stop conditions.

Assessment requirements

- Associate each option with the prompt programmatically.
- Provide visible focus indicators.
- Do not impose a speeded time limit; listed times are estimates.
- Announce correctness and rationale to screen readers after submission.
- Make JSON content available through an accessible rendering layer rather than requiring learners to read raw JSON.

Troubleshooting and stop conditions

Problem	Instructor response
MW endpoint fails	Display failure evidence, use only the versioned cache, label it cached, and schedule a later access recheck.
Live count differs from guide	Check revision, endpoint schema, selected analyses, QC roles, and parsing; document the difference rather than forcing the old count.
QPP... rows appear in biological PCA	Stop, correct sample_role , rerun participant/sample counts, then refit PCA.
Exercise matrix appears only as blood	Inspect collection metadata; preserve the broad factor and derive serum from the specific supported source.
Crosswalk join multiplies rows	Stop and audit key cardinality; do not drop duplicates randomly.
RefMet mapping maximizes overlap by narrowing ambiguous labels	Restore source detail, mark review required or broader, and recompute eligible sets.
Learner proposes pooled diet-versus-exercise inference	Return to the estimand and confounding audit; require a within-study or descriptive name-set alternative.
Access policy is unclear	Pause affected movement, document the question, and use the resource's official support channel.
Learner lacks a working Python environment	Use the first-time setup guide and default cache. An instructor demonstration may preview the output, but schedule the learner's hands-on run after setup rather than treating the demonstration as completion.

Quality and fidelity audit

Before declaring the module complete, inspect evidence for every requirement:

- Five lessons total exactly 140 estimated minutes.
- Each lesson has objectives, narrative, a worked example, activity, knowledge check, guardrails, accessibility support where visual content is proposed, and primary sources.
- The locked pair and reserve pair are recorded.
- Every claim about access has current first-party evidence and a check date.
- Pretest, posttest, and embedded checks cover all five objectives and include answer rationales.
- Learner and instructor versions exist for all three templates.
- The overlap audit distinguishes 153 raw from 145 conservative biological names.

- The lesson never claims that harmonization establishes quantitative pooling, causal comparability, or shared biological response.
- Generated dashboard, notebook, and support exports render accessibly and preserve all warnings.

Primary references

1. Metabolomics Workbench. [REST Service](#), [mwTab documentation](#), and [RefMet](#). Accessed August 10, 2026.
2. Fahy E, Subramaniam S. [RefMet: a reference nomenclature for metabolomics](#). *Nature Methods*. 2020;17:1173-1174.
3. MoTrPAC Study Group, Jakicic JM, Kohrt WM, et al. [MoTrPAC human studies design and protocol](#). *Journal of Applied Physiology*. 2024;137(3):473-493.
4. NIH Common Fund. [Nutrition for Precision Health FAQ](#). Accessed August 10, 2026.
5. All of Us Research Program. [Researcher Workbench](#) and [policy guidance](#). Accessed August 10, 2026.
6. Wilkinson MD, Dumontier M, Aalbersberg IJ, et al. [The FAIR Guiding Principles](#). *Scientific Data*. 2016;3:160018.

Metabo-Diet answer key

Instructor-only file: Do not expose before learners submit the corresponding assessment. The JSON assessment files are the authoritative machine-readable source for item wording, options, answers, and full rationales.

Pretest key

Item	Answer	Objective(s)	Rationale
PRE-01	C	LO1	A post value needs an anchor, elapsed time, physiological state, specimen, and assay context.
PRE-02	B	LO1, LO2	Repeated samples from one person are correlated; sample count is not participant count.
PRE-03	A	LO2	A shared standardized name supports a naming bridge, not shared scale, specimen, preprocessing, or causal response.
PRE-04	B	LO3	RefMet supplies analytical-chemistry-centered standardized nomenclature.
PRE-05	C	LO3	Ambiguity must remain visible until additional evidence supports one candidate.
PRE-06	B	LO4	The MW study-context factors output returns samples and experimental variables.
PRE-07	B	LO2, LO4	PCA describes variance under selected preprocessing and does not assign a cause.
PRE-08	B	LO5	Access is established with current first-party evidence for the exact release and action.
PRE-09	B	LO1, LO2	Plasma and serum can share a broad discovery category but remain different matrices.
PRE-10	B	LO5	Participant-level governed data stay in approved compute and storage; only permitted outputs leave.

Maximum: 10 points. Use as a baseline, not a pass/fail gate.

Embedded knowledge-check key

Item	Answer	Key idea
KC1-01	B	RefMet match is a candidate standardized naming bridge, not analytical equivalence.
KC1-02	C	Verify exact dataset, release, action, and current first-party policy.
KC1-03	C	Cross-study PCA separation is not uniquely attributable when study and phenotype are confounded.

Item	Answer	Key idea
KC2-01	B	A coarser binary derivation can be partially comparable with documented information loss.
KC2-02	C	Repeated measures require a documented subject-sample relationship.
KC2-03	B	A physiological timepoint needs anchor, offset, and state.
KC3-01	C	Preserve multiple candidates and exclude unresolved rows from exact overlap.
KC3-02	B	Unit conversion also requires compatible quantity, matrix, denominator, and quantification basis.
KC3-03	B	Reproducibility requires source, mapping, review, rationale, and purpose-specific eligibility.
KC4-01	A	QPP... rows are pooled QC candidates, not participant-timepoint observations.
KC4-02	B	153 is the raw nonblank exact RefMet string intersection before exclusions; the conservative artifact-filtered count is 145.
KC4-03	A	Separate within-study PCA is exploratory and remains sensitive to preprocessing and design.
KC5-01	C	MoTrPAC currently has public and restricted paths depending on the release.
KC5-02	A	Approved code and permitted public references move in; controlled participant-level data do not move out.
KC5-03	B	Access feasibility and scientific compatibility are separate gates.

Give feedback immediately after each embedded response and permit a retry. Record first attempts only if the delivery and evaluation plan permits.

Posttest key

Item	Answer	Objective(s)	Rationale
POST-01	B	LO1	The current ST001521 factor table has 150 participant-timepoint rows and 10 pooled QPP... QC rows.
POST-02	B	LO1, LO2	Preserve plasma and serum and limit the broad blood-derived category to discovery.
POST-03	B	LO1, LO2	FARMM Day 9 and 3-hour post-exercise recovery have different anchors and protocol meaning.
POST-04	B	LO2	Study-specific population, matrix, lab, platform, time, and protocol differences confound a direct contrast.

Item	Answer	Objective(s)	Rationale
POST-05	B	LO3	Isotope/internal-standard roles must be preserved and excluded from biological overlap.
POST-06	B	LO3	Preserve feature-level provenance; a set collapse is distinct from a quantitative feature merge.
POST-07	B	LO3, LO4	The audited flow is 153 raw nonblank exact strings and 145 after eight nonbiological ST003348 exclusions.
POST-08	A	LO2, LO4	Quantitative exploration is fitted within study; the uncalibrated raw matrices are not stacked.
POST-09	B	LO4	Cache use requires failure evidence, timestamp, integrity verification, and a visible source flag.
POST-10	C	LO5	MoTrPAC access is dataset- and release-dependent and must be checked.
POST-11	B	LO5	NPH participant-level analysis belongs in the Researcher Workbench and output follows current policy.
POST-12	C	LO3, LO4	Unexpected row multiplication requires a join-cardinality audit before analysis.

Maximum: 12 points. Suggested mastery is 10/12 plus satisfactory performance artifacts. A learner who misses both items in an objective domain should review that lesson and complete a parallel retry.

Lesson 4 interpretation challenge

Statement 1

Original: "The studies share 153 biologically identical metabolites."

Classification: Unsupported.

Suggested revision:

The audited build found 153 shared nonblank exact RefMet strings before exclusions and 145 conservative biological overlaps after removing eight [ST003348](#) isotope/internal-standard rows; these are name-set results, not evidence of identical concentration or response.

Statement 2

Original: "The diet study contains 160 participants."

Classification: Unsupported.

Suggested revision:

[ST001521](#) describes 30 participants and the current factors response contains 150 participant-timepoint rows plus 10 pooled QPP... QC rows.

Statement 3

Original: "The exercise samples are blood, so they are the same matrix as diet plasma."

Classification: Unsupported.

Suggested revision:

The exercise factors use the broad label blood, while collection metadata specifies serum; serum and diet-study plasma are partially comparable as blood-derived fluids for discovery and are not the same quantitative matrix.

Statement 4

Original: "PCA separates exercise timepoints, proving the race walk caused every measured change."

Classification: Unsupported.

Suggested revision:

Within ST003348, the exploratory PCA may show variation associated with the ordered REST, STAT, REC3, and REC22 samples under the chosen preprocessing; repeated measures, collection context, technical variation, and other time-varying factors limit causal attribution for individual metabolites.

Statement 5

Original: "An apparent class difference may reflect assay coverage as well as biology."

Classification: Supported, with a precision edit.

Suggested revision:

A difference in the number or proportion of reported RefMet names by class may reflect platform coverage, annotation, filtering, matrix, and biology; it should be described as deposited analyte coverage rather than class abundance.

Cohort-comparison worksheet key points

Use the detailed instructor worksheet for row-level evidence. Require these conclusions:

- Both studies are human, which is directly comparable only at a broad species-description level.
- ST001521 is plasma and ST003348 is serum. The ST003348 factor source says blood, but collection metadata provides the supported specific matrix.
- ST001521 includes five longitudinal study times in a feeding plus Abx/PEG protocol; ST003348 includes an acute exercise and recovery sequence. They do not form one common post-intervention axis.
- ST001521 has 30 described participants, 150 biological rows, and 10 pooled QC rows; ST003348 has 19 represented athletes and 76 biological rows.

- The diet study has a hybrid design: established vegan outpatients and randomization of 20 omnivores to omnivore versus EEN. Do not call all three groups randomized.
- Analysis families and some ion modes overlap, but laboratory, chromatography, matrix, coverage, preprocessing, and scale differ.
- Cross-study quantitative pooling is not authorized.

Model exclusion statement

We will not combine peak-area values across the studies because matrix, laboratory, chromatography, feature coverage, preprocessing, population, time, and intervention context differ. We can still compare reviewed RefMet names by building one documented set for each study and reporting their overlap.

Metabolite/metadata crosswalk key points

Use the 20-point rubric in the instructor template. Minimum acceptable evidence includes:

- Original label, source RefMet value, accession, analysis ID, and endpoint.
- Mapping and annotation-resolution status.
- Reviewer or rule version and decision rationale.
- An ambiguous or broader mapping kept visible.
- An unmapped row retained.
- A many-to-one mapping retained at feature level.
- Isotope/internal-standard rows excluded from biological overlap with a reason.
- Separate eligibility for exact overlap, class summary, within-study analysis, and cross-study quantitative analysis.
- `eligible_cross_study_quantitative = no` throughout.
- Audit flow that reproduces or explains 153 raw and 145 conservative overlaps.

Access-pattern transfer checklist key points

Use the 20-point rubric in the instructor template. Require:

- Exact dataset, release, data level, intended action, compute, storage, and outputs.
- Current first-party evidence with a check date.
- Separate access and scientific-compatibility decisions.
- An explicit data-flow boundary.
- No credentials or governed data in the artifact.

- A [GO](#), [REVISE](#), [WAIT](#), or [STOP](#) decision with unresolved items.

MoTrPAC worked conclusion

Mixed: current first-party documentation distinguishes public releases accessible without login from restricted or embargoed content requiring additional authorization. Verify the exact file and intended action.

NPH/All of Us worked conclusion

Controlled compute for participant-level work: run approved code in the Researcher Workbench, keep governed data there, and export only outputs permitted by current policy. Verify the exact Registered or Controlled Tier collection and release.

Performance-artifact minimums

A learner should not pass the module based on multiple-choice score alone. Require:

- Comparison worksheet with sources, explicit missingness, and at least one intentional non-harmonization decision.
- Crosswalk score of at least 16/20 with all scientific-safety conditions met.
- Notebook audit that reproduces the expected structural checks and either reproduces 153/145 or explains a documented data revision.
- Four-sentence interpretation containing observation, context, alternatives, and a claim boundary.
- Transfer checklist score of at least 16/20 with no governance-safety failure.

Cohort-comparison worksheet - instructor version

This version supplies expected evidence and defensible classifications for the locked August 10, 2026 build. Repository records can change; learners should receive credit for a different value when they provide current, traceable evidence and update the reasoning appropriately.

1. Comparison purpose

Recommended purpose:

Describe exact reviewed RefMet-name overlap and broad chemical-class coverage across the two deposited studies while retaining population, plasma-versus-serum, time, platform, and scale differences.

This purpose does not authorize pooling peak areas or estimating a diet-versus-exercise effect.

2. Dataset identity and access evidence

Field	Diet study	Exercise study
Configuration	DIET_ACCESSION = ST001 521	EXERCISE_ACCESSION = ST003 348
Title	Plasma metabolites of known identity profiled using hybrid nontargeted methods (part-III)	An integrated LC-MS analysis of the biometric characteristics of different time cohorts of race walkers - untargeted
Study DOI	Project DOI 10.21228/M8B984	Project DOI 10.21228/M8C802
Landing page	https://www.metabolomicsworkbench.org/data/DRCCMetadata.php?StudyID=ST001 521	https://www.metabolomicsworkbench.org/data/DRCCMetadata.php?StudyID=ST003 348
REST summary	/rest/study/study_id/ST001 521/summary	/rest/study/study_id/ST003 348/summary
License in summary	CC BY 4.0	CC BY 4.0
Public status at content build	Public REST responses retrieved	Public REST responses retrieved

The learner must add their own UTC check time and live/cached status.

3. Design and population key

Field	ST001521 expected evidence	ST003348 expected evidence	Defensible classification
Species	Homo sapiens	Homo sapiens	D for species-level description
Design	Longitudinal FARM feeding protocol; three diet groups; concurrent Abx/PEG perturbation; mixed assignment/setting	Repeated measures across one endurance exercise bout and recovery	P for repeated-measures structure; N as one common intervention design

Field	ST001521 expected evidence	ST003348 expected evidence	Defensible classification
Participants	30 healthy volunteers described; 10 per diet group	20 recruited male student-athletes described in collection metadata, one withdrew; 19 represented in 76 samples	P for descriptive counts only
Biological samples	150 expected participant-timepoint rows, 30 x 5	76, 19 x 4	P ; do not treat as independent people
Other samples	10 QPP... pooled QC candidates among 160 factor rows	No additional QC rows identified in the factor endpoint	N as participant data; retain for QC role
Repeated linkage	Local identifiers support subject and visit parsing, to be validated	Local identifiers such as 79_1 through 79_4 support athlete and time parsing, to be validated	P until parsing is checked against documentation
Assignment caveat	Twenty omnivores randomized to omnivore versus EEN; vegan participants were established vegans and continued usual diet as outpatients	No randomized comparison group in the deposited acute time-course description	N for a common randomized contrast

High-quality learner answers note that the project metadata labels [ST001521](#) as observational while its detailed summary includes randomization within the 20 omnivores. The resolution is not to choose one label, but to describe the hybrid design at the relevant contrast.

4. Phenotype and protocol key

Field	ST001521	ST003348	Decision
Anchor	Diet group plus longitudinal microbiota depletion/reconstitution context	Acute endurance race walk and recovery	N as one intervention variable
Groups/time	Deposited Study_Diet values include Vegan , Western , and Modulen ; time values include Baseline and Days 5, 9, 12, 15	REST , STAT , REC3 , REC22	N as a common time axis
Concurrent procedures	Vancomycin/neomycin days 6-8 and PEG on day 7 described in summary	Exercise plus recovery; fasting noted for REST and REC22 in collection summary	N for diet-only versus exercise-only attribution
Setting	Vegan outpatients; omnivore/EEN inpatient under supervision	Athlete exercise study; exact supervision details should be sourced	N or P depending purpose

Learners should not describe all three diet groups as randomized.

5. Specimen and pre-analytics key

Field	ST001521	ST003348	Decision
Factor label	Blood (plasma)	blood	Source labels retained
Specific matrix	Plasma, isolated after blood collection	Collection block specifies Blood (serum) and clotting before centrifugation	P for blood-derived-fluid discovery; N for treating matrix as identical
Storage	Plasma aliquots described as immediately frozen at -80 C	Sera frozen on dry ice and stored at -80 C	P; other pre-analytics still differ
Collection timing	Study days in a feeding and Abx/PEG protocol	REST at 8:20 am, STAT at 10:30 am, REC3 at 13:30, REC22 at 8:20 am next day per collection summary	N as common clock/time protocol

The factors-versus-collection difference for ST003348 belongs in the provenance-conflict table. A good resolution preserves blood as the source factor, derives serum from the more specific collection metadata, and cites both.

6. Timepoint key

ST001521

Source	Canonical interpretation
Baseline	Pre-protocol or initial collection; verify exact day mapping in the study record
Day 5	Longitudinal study day before the documented Abx/PEG interval
Day 9	Longitudinal study day after the documented days 6-8 antibiotics and day 7 PEG
Day 12	Later study day in reconstitution/feeding context
Day 15	Later study day in reconstitution/feeding context

The labels are valid within the FARMM design but should not be converted to acute recovery offsets.

ST003348

Source	Canonical interpretation
REST	Pre-exercise fasting sample
STAT	Immediately after exercise
REC3	3-hour recovery
REC22	22-hour recovery, fasting sample on day 2

The two time axes are **N** for a common post-intervention variable. A broad **within-study repeated time** flag is **P** for workflow comparison only.

7. Assay key

ST001521

Analysis	Summary	Chromatography	Instrument	Ion mode	Units
AN002533	HILIC	HILIC	Thermo Exactive Plus Orbitrap	Positive	Unitless peak areas
AN002534	Reversed phase	Reversed phase	Thermo Exactive Plus Orbitrap	Positive	Unitless peak areas
AN002535	HILIC	HILIC	Thermo Q Exactive Plus Orbitrap	Negative	Unitless peak areas
AN002536	Reversed phase	Reversed phase	Thermo Q Exactive Orbitrap	Negative	Unitless peak areas

ST003348

Analysis	Summary	Chromatography	Instrument	Ion mode	Units
AN005483	Reversed phase	Reversed phase	Thermo Q Exactive Plus Orbitrap	Positive	Peak area
AN005484	Reversed phase	Reversed phase	Thermo Q Exactive Plus Orbitrap	Negative	Peak area

Some instrument families and ion modes overlap, but laboratories, chromatography, specimen, feature coverage, preprocessing, and measurement scales still differ. The correct cross-study quantitative pooling decision is **No** for this module.

8. Minimum comparability decisions

Domain	Purpose	Expected category	Allowed action	Prohibited inference
Species	Cohort description	D	State both are human	Assume populations are exchangeable
Matrix	Broad discovery	P	Label blood-derived plasma versus serum	Treat values as same matrix
Time	Within-study trajectory	P	Analyze each study's time course separately	Map Days 5-15 to STAT/REC3/REC22
Phenotype	Metadata comparison	N as one field	Describe diet and exercise protocol structures	Estimate a common intervention effect
RefMet name	Exact reviewed overlap	P until annotation QC; then D at name-set level for accepted rows	Count eligible shared names	Infer equal concentration or response
Units	Quantitative cross-study comparison	N	Report source scales	Pool unitless/peak areas

Domain	Purpose	Expected category	Allowed action	Prohibited inference
Repeated measures	Workflow structure	P	Preserve subject linkage within each study	Treat rows as independent participants
Platform/mode	Assay inventory	P	Stratify or document modes	Assume equal coverage because both use LC-MS

9. Model exclusion statement

We will not combine peak-area measurements across the two studies because matrix, laboratory, chromatography, feature coverage, preprocessing, scale, population, and intervention context differ. We can still compare reviewed RefMet-name coverage by constructing provenance-preserving study-specific sets and reporting their intersection with explicit eligibility rules.

10. Acceptable bounded claim

Under the specified mapping and eligibility rules, both public studies report a set of shared standardized metabolite names; this overlap describes deposited analyte coverage and does not establish shared abundance or intervention response.

Sources for instructors

- [ST001521 landing page](#)
- [ST001521 factors](#)
- [ST001521 analyses](#)
- [ST003348 landing page](#)
- [ST003348 factors](#)
- [ST003348 analyses](#)

Metabolite and metadata crosswalk - instructor version

Use this key to model decision quality. It is not a substitute for the machine-generated full crosswalk, whose current counts and rows must be created from the versioned repository responses.

1. What earns full credit

A full-credit crosswalk:

- Preserves accession, analysis ID, source identifier, submitted name, and source RefMet result.
- Records exact endpoints and retrieval dates.
- Treats mapping as a reviewable decision rather than a truth assignment.
- Preserves annotation resolution and identification evidence.
- Retains ambiguous, unmapped, isotope-labeled, and duplicated cases.
- Separates exact-name, class-summary, within-study, and cross-study eligibility.
- Marks cross-study quantitative eligibility **no** for every row in this module.
- Produces an overlap flow with exclusion counts, not only an intersection.

2. Completed metabolite examples

These examples use values visible in the Metabolomics Workbench metabolite responses. Review status is an instructional decision based on the visible naming evidence; a production crosswalk should incorporate all available identifiers and methods.

Source	Analyses	Deposited name	Source RefMet	Recommended status	Exact overlap	Class summary	Key reason
ST001521	AN002533	2-Aminooctanoic acid	2-Aminocaprylic acid	review_required until synonym/identifier evidence is checked; then potentially accepted_exact	review	review/yes	Name changed to a synonym; verify it represents the same supported structure
ST001521	AN002533	LPC(P-18:0)/LPC(O-18:1)_A	LPC P-18:0 or LPC O-18:1	accepted_broader at the explicitly ambiguous species level	no for a unique molecular structure; yes only if exact key is the ambiguous RefMet string	yes	The source does not distinguish plasmalogen and plasmalogen alternatives
ST001521	AN002533	LPE(18:0)_A	LPE 18:0	accepted_exact at reported sum-composition level, with duplicate flag	yes as a set key	yes	_A and another source feature can map to one RefMet name; do not merge intensities silently
ST001521	AN002533	LPE(18:0)_B	LPE 18:0	accepted_exact at reported sum-composition level, with duplicate flag	yes as same set key	yes	Many-to-one mapping must remain visible

Source	Analyses	Deposited name	Source RefMet	Recommended status	Exact overlap	Class summary	Key reason
ST003348	AN005484	Mannitol+Sorbitol	Mannitol	review_required or accepted_broader only for a stated broad purpose	no	review	Composite/isomeric label has been narrowed by the returned name; information loss is material
ST003348	AN005484	NegX-RT295MZ165	blank	unmapped	no	no/unclassified only if rule allows	Unidentified feature must not be assigned from mass alone
ST003348	AN005484	Palmitic acid-[13C]16	Palmitic acid	excluded_nonbiological for biological overlap	no	no	Isotope-labeled analyte/standard must not create a biological overlap with endogenous palmitic acid
ST003348	AN005484	Linoleic acid(FFA(18:2 n6))	Linoleic acid	review_required, then potentially accepted at supported resolution	review	yes/review	Preserve source notation and verify structural-resolution equivalence

Important: the exact handling of A/B features depends on source-method context. They may be chromatographic isomers or distinct signals. Set overlap can collapse identical accepted keys while the quantitative matrix retains source features.

3. Completed metadata examples

Source field/value	Harmonized representation	Compatibility	Purpose	Decision note
ST001521: sample_source = Blood (plasma)	broad_source = blood_derived_fluid; matrix = plasma	Partial with exercise specimen	Name-set discovery	Preserve plasma; broad category is not quantitative equivalence
ST003348 factors: sample_source = blood; collection: Sample Type = Blood (serum)	source_factor = blood; matrix = serum; matrix_evidence = collection	Partial with diet specimen	Name-set discovery	Preserve both sources; prefer the specific supported matrix for analysis
ST001521: Time = Day 9	study_day = 9; protocol_state = post_Abx_PEG_interval if supported by summary	Not comparable with REC time	Within-study description	Do not call it generic post or map to exercise recovery
ST003348: Collection_time = stat	anchor = exercise_end; offset = 0 min; state = immediate_post_exercise	Not comparable with diet days	Exercise trajectory	Meaning comes from project/collection summary, not the code alone

Source field/value	Harmonized representation	Compatibility	Purpose	Decision note
ST003 348: Collection_time = rec3	anchor = exercise_end; offset = 3 h; state = recovery	Direct only inside exercise study	Exercise trajectory	Preserve REC3 source label
ST001 521: QPP01 with diet/sex/time NA	sample_role = pooled_qc after evidence check	Not a participant observation	QC diagnostics	Exclude from biological PCA and participant counts; retain for QC
ST001 521: units = unitless peak areas	Keep source scale; no common target	Not compatible for pooled quantitative analysis	Measurement audit	Scaling cannot create a calibrated common unit
ST003 348: units = Peak area	Keep source scale; no common target	Not compatible for pooled quantitative analysis	Measurement audit	Different assay, lab, matrix, and preprocessing remain

4. Overlap audit key

The build-level checks are 153 shared nonblank exact RefMet strings before artifact and eligibility exclusions, then 145 conservative biological overlaps after excluding eight ST003 348 isotope/internal-standard rows that map to ordinary RefMet labels. A full-credit learner response:

1. Labels this as preliminary.
2. Records the retrieval and crosswalk versions.
3. Shows how isotope-labeled compounds, artifacts, blanks, duplicates, ambiguous mappings, and annotation rules change the set.
4. Reports diet, exercise, and shared denominators.
5. Does not describe shared strings as shared intervention responses or comparable concentrations.

Require learners to distinguish the 153 raw check from the 145 conservative artifact-filtered check. Further justified eligibility rules may reduce the reported set below 145; they must be versioned and counted.

5. Common errors and feedback

Error	Instructor feedback
Replacing source names with RefMet names	Add separate source and standardized columns so the mapping remains auditable.
Accepting every nonblank RefMet value	A repository-provided mapping still needs role, resolution, artifact, and ambiguity review.
Dropping blank mappings	Retain them as unmapped; missing mapping is a meaningful result.

Error	Instructor feedback
Mapping isotope-labeled standards to endogenous names	Preserve isotope labels and exclude nonbiological standards from biological overlap.
Merging duplicate RefMet features before preserving analysis IDs	Keep feature-level rows; create a separate set-level key for overlap.
Marking plasma and serum identical	Retain exact matrix and use a broad category only for a purpose that tolerates the distinction.
Converting peak areas to a common z-score and calling them comparable	Standardization changes scale but does not remove platform, matrix, batch, or design differences.
Maximizing the mapping rate	Mapping quality and explicit uncertainty matter more than the percentage mapped.

6. Suggested scoring, 20 points

Criterion	Points
Source identity and provenance complete	4
Mapping and annotation-resolution decisions justified	4
Ambiguous, unmapped, isotope-labeled, and duplicate cases handled	4
Specimen/time/unit metadata aligned with information loss visible	4
Purpose-specific eligibility and non-pooling guardrail applied	4

Access-pattern and transfer checklist - instructor version

This key provides two worked cases and a scoring rubric. Access rules change; instructors must recheck all linked first-party documentation immediately before delivery. A learner with newer documented evidence should not be penalized for a different conclusion.

Core rule

The unit of assessment is:

specific dataset + release + data level + intended action + user/institution +
compute/storage/output plan + check date

Answers that label an entire program [open](#), [gated](#), or [controlled](#) without those qualifiers are incomplete.

Worked case A - MoTrPAC

Scenario

The learner wants to download a public MoTrPAC analysis-result file for local, noncommercial exploratory work and compare its metadata structure with [ST003348](#).

Evidence expected as of August 10, 2026

- The MoTrPAC FAQ states that public data are available through the portal and that public data are accessible without login, while restricted data require authentication.
- The Data Hub documentation describes public releases and also an embargoed limited acute exercise rat subset requiring an account and DUA.
- Consortium-member onboarding explicitly says it does not apply to community users accessing public data.

Primary pages:

- [MoTrPAC FAQ](#)
- [MoTrPAC Data Hub](#)
- [MoTrPAC onboarding](#)

Correct planning classification

[Mixed](#), with the exact public file potentially following open public retrieval. Restricted or embargoed files follow an agreement/approval-gated path. The learner must still record license/terms for the selected public file.

Architecture

For a verified public file, local analysis and a public cache may be possible under the applicable terms. Replace the MW REST retrieval adapter with the documented MoTrPAC download method. Retain provenance, crosswalk, sample-role, and scientific-compatibility checks.

For a restricted file, the learner must obtain authorization, honor storage and publication rules, and exclude restricted content from module caches and public repositories.

Scientific decision

Access does not establish compatibility. Compare exercise modality, tissue, participant structure, timepoint, assay, preprocessing, and scale before any integration. A MoTrPAC exercise dataset is not automatically comparable with the race-walking serum data.

Worked case B - NPH/All of Us

Scenario

The learner wants to adapt the crosswalk and within-study analysis to a specified NPH release in the All of Us Researcher Workbench.

Evidence expected as of August 10, 2026

- The NIH NPH FAQ says NPH data are stored in and analyzed within the All of Us Researcher Workbench and that the platform is available to approved researchers.
- All of Us documentation describes Registered and Controlled Tier collections and governed access requirements.
- Current support documentation prohibits participant-level data and direct or inferable small counts outside the Workbench and provides dissemination guidance for allowable outputs.

Primary pages:

- [NPH FAQ](#)
- [All of Us Researcher Workbench](#)
- [All of Us policy questions](#)
- [All of Us access-tier overview](#)

Correct planning classification

[Controlled compute](#) for participant-level analysis, with the exact Registered or Controlled Tier and NPH release verified. Do not claim that no outputs can ever leave; state that only policy-permitted outputs may leave under current dissemination rules.

Architecture

- Move approved analysis code and permitted public reference resources into the Workbench.
- Keep participant-level and restricted derived data inside approved storage.
- Do not populate the module's local cache with NPH participant data.
- Use a dated RefMet snapshot if live external calls are unavailable or inappropriate.
- Review plots, counts, tables, code outputs, and model artifacts against current policy before export.
- Keep secrets out of code, notebooks, logs, and repositories.

Scientific decision

NPH may support richer diet-response questions, but each module, diet period, test meal, specimen, assay, repeated measure, and target estimand must be defined. Controlled access does not make a study scientifically suitable, and a richer dataset does not justify a vague question.

Common misconceptions

Misconception	Corrective feedback
"MoTrPAC is registration-gated."	Current documentation shows both public and restricted paths. Assess the exact release and action.
"MoTrPAC is open."	Some releases are public, while restricted or embargoed content can require authentication and a DUA.
"NPH data can never produce an external result."	Participant-level data remain governed, but policy-permitted aggregate outputs can be disseminated. Verify current rules.
"A public landing page means the data can be downloaded."	Discovery and download are separate actions; document each.
"I have approval, so I can put data in GitHub."	Authorization is bounded by storage, sharing, and output rules.
"Access is the only issue."	A dataset can be accessible and still be unsuitable for the intended scientific comparison.

Suggested scoring, 20 points

Criterion	Points
Exact dataset, release, action, and intended flow scoped	3
Current first-party evidence and check dates recorded	4
Account/agreement/compute/output rules correctly separated	4

Criterion	Points
Scientific compatibility assessed independently	3
Workflow and data-flow adaptations are concrete and compliant	4
Decision and unresolved dependencies are explicit	2

Minimum pass conditions

Regardless of numeric score, require revision if the learner:

- Places credentials or sensitive identifiers in the worksheet.
- Proposes exporting participant-level governed data without authority.
- Uses a program-wide access label without checking the selected release.
- Treats access approval as scientific compatibility.
- Proposes local or public caching of controlled participant data.

Appendix - Data provenance

This appendix summarizes the versioned evidence trail used by the release. The machine-readable source is `module/data/provenance.json`.

Provenance field	Release value
Repository	Metabolomics Workbench / National Metabolomics Data Repository
Retrieval time (UTC)	2026-08-10T11:44:11Z
Access model	open; no login, registration, credential, or data-use agreement required for the selected public REST records
Selection status	locked
Manifest schema	1.0.0

Locked study pair

Role	Accession	Specimen	Design summary	License
Diet	ST001521	Blood (plasma)	FARMM longitudinal diet study. Twenty omnivores were randomized to inpatient Western/omnivore or exclusive enteral nutrition (Modulen) feeding; ten established vegans continued their usual diet as outpatients. Plasma was collected at baseline and days 5, 9, 12, and 15.	CC BY 4.0
Exercise	ST003348	Blood (serum)	Acute endurance race-walking study with fasting pre-exercise REST, immediate post-exercise STAT, 3-hour recovery REC3, and fasting 22-hour recovery REC22 serum samples.	CC BY 4.0

RefMet overlap audit

Measure	Value
Diet unique nonblank RefMet names	510
Exercise unique nonblank RefMet names	475
Raw exact overlap	153
Known isotope/internal-standard exclusions	8
Conservative biological overlap	145

The 145-name count excludes eight known labeled/internal-standard mappings and is the operational biological crosswalk count. Further annotation curation may reduce it. Overlap supports identifier-

crosswalk and class-summary exercises; it does not make raw peak areas quantitatively comparable across studies.

Sanitization and claim boundaries

Files in module/data/raw are exact source snapshots and must not be edited in place. Sanitized/derived tables must retain source IDs and record these transformations.

St001521 Factors

- Trim factor-field whitespace without changing repository labels.
- Exclude QPP01 through QPP10 and rows whose Study_Diet, Sex, and Time factors are NA from biological analyses; retain them only as pooled-QC records.
- Derive biological participant_id from the leading four digits of local_sample_id and verify observed visits rather than forcing a balanced panel.
- Preserve Western as the factor label and document that the narrative calls this arm omnivore.
- Use the valid split study REST endpoints; do not ingest malformed AN002533, AN002535, or AN002536 mwTab JSON.

St003348 Factors

- Derive participant_id from the integer before the underscore in local_sample_id.
- Validate suffixes 1, 2, 3, and 4 against rest, stat, rec3, and rec22, respectively.
- Retain source sample_source=blood while setting harmonized specimen to serum from the collection block.
- Treat the treatment-block value no treatment as a metadata defect; define the exercise contrast from the documented protocol, collection, and time factors.

Cross Study Metabolites

- Remove the eight explicit isotope-labeled/internal-standard mappings listed in primary_pair.overlap before constructing the biological RefMet crosswalk.
- Keep analysis_id attached to duplicated RefMet names unless a documented within-study collapse rule is applied.
- Never concatenate uncalibrated peak areas across the plasma and serum studies.

Cached-source integrity

Study	Cached file	Bytes	SHA-256
ST001521	ST001521_summary.json	575	d2199dc631f33c09b9d9cb13f64a9c5233e1b6b7fe3f372e4cb8e947817541d9

Study	Cached file	Bytes	SHA-256
ST001521	ST001521_factors.json	49194	7b88427c7434f20538aff398f8ddac7e88d666696f4e19992c85d6017fe0372f
ST001521	ST001521_analysis.json	2025	72d60232e8a0f39eb24c790cabe7bd1585a1e852eb58f4b0288703273a0c70ac
ST001521	ST001521_metabolites.json	95806	7686f09d81f632d3996d5a5eceaedf31632a3c160d291f21cfa574d680251a53
ST001521	ST001521_data.json	1997104	5528f4a721189f45de65d65b325727177bc5355181f571827f6030a7e7676c06
ST003348	ST003348_summary.json	621	3766073c903b8975c2959fbb9c562b4cee531cddc02e749f380915354b193bee
ST003348	ST003348_factors.json	11791	e31bb851873fee1d185a2d91d8b5553637145d799793d416ecf2be6c3f8c3329
ST003348	ST003348_analysis.json	1051	27f654c3cee086b2310ebe4aeee64f5a835949e99c8b0da4736e6316652c4a19
ST003348	ST003348_metabolites.json	106651	492be3dd45c6bd741b64634249e4acc89a3d0c9750f98f816215305cff5ef566
ST003348	ST003348_data.json	1076605	0610a28e0e858524adefee5ee2d0dca46cf75c9a5bab1ebcb659e2573715adfb
RefMet	refmet_classification.json	31375639	b11b6f09c5f068226ac440b815c693af0255e506a2cb33736a265004a95cb50d

Data attribution and reuse

Primary public datasets

This training package includes cached copies and derived educational artifacts from two public Metabolomics Workbench/NMDR studies. Each study's public summary endpoint reports a Creative Commons Attribution 4.0 International license.

Diet-anchored case study

- Metabolomics Workbench study: [ST001521](#)
- Project: [PR001024](#)
- Project DOI: [10.21228/M8B984](#)
- Repository record: [Metabolomics Workbench ST001521](#)
- License: [CC BY 4.0](#)

Exercise-anchored case study

- Metabolomics Workbench study: [ST003348](#)
- Project: [PR002083](#)
- Project DOI: [10.21228/M8C802](#)
- Repository record: [Metabolomics Workbench ST003348](#)
- License: [CC BY 4.0](#)

RefMet classification

- Source: Metabolomics Workbench RefMet REST service
- Endpoint: [RefMet classification REST endpoint](#)
- Documentation: [Metabolomics Workbench REST documentation](#)

Exact retrieval times, response sizes, SHA-256 checksums, release fields, and transformations are recorded in [module/data/provenance.json](#) in the repository and analysis bundle.

Training materials

Except for third-party material identified here or in [module/data/provenance.json](#), the lesson text, original templates, notebook narrative and code, scripts, figures, and assessments in this package are licensed under the [Creative Commons Attribution 4.0 International Public License](#). Reusers must give appropriate credit, link to the license, and indicate changes. The repository and release packages include the full English legal code; it is also available from [Creative Commons](#).

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Interpretation notice

The package uses these studies to teach metadata reasoning, identifier harmonization, and within-study exploratory analysis. It does not treat the studies as one experiment, estimate a diet-versus-exercise causal effect, or claim that plasma and serum peak areas are quantitatively interchangeable.